

Blockdrive: A Secure, Transparent, and Zero Web3 Friction n-Tier Architecture for Verifiable Credentials

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Abstract

Web2 credentialing services carry privacy risks from centralized databases held by individual issuers, while Web3 solutions suffer from low transaction rates. We propose Blockdrive, an n-tier architecture utilizing asynchronous Go orchestrators for 50x higher throughput than traditional Web3 solutions. Combining the high-speed Sonic chain and Encrypt-then-Pin IPFS storage, it delivers a frictionless Web2 experience with uncompromising Web3 security.

CCS Concepts

• **Software and its engineering** → **n-tier architectures**.

Keywords

System Architecture, Web3, Asynchronous Processing, Soul-bound Tokens

1 Introduction

Enterprise Web3 adoption often stalls on RPC timeouts, wallet complexity, and gas fees. Blockdrive bypasses these operational hurdles via decoupled front-ends and OAuth-mapped Shadow Users with automated gas sponsorship.

2 System Architecture

Blockdrive’s multi-tiered architecture (Figure 1) strictly partitions components to maximize performance and security.

Front-End (Next.js): Issues async *API Requests* for bulk issuance to avoid finality delays. Clients fetch credentials via IPFS CIDs through a dedicated gateway.

Orchestrator (Go): Offloads requests to background queues, preventing timeouts. It synchronizes PostgreSQL states, mapping OAuth identities to Web3 addresses (Shadow User) seamlessly bridging Web2 and Web3.

Hybrid Storage: Encrypts PDFs via AES-256 before an *Upload* to IPFS, ensuring decentralized payloads remain undecipherable without authorized off-chain keys.

Blockchain (Sonic): Mints an ERC-5192 Soulbound Token (SBT) embedding the secure CID, finalizing the trustless issuance of immutable, non-transferable records.

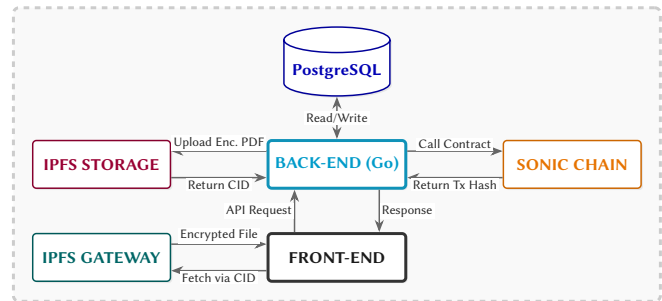


Figure 1: Decoupled Orchestration Flow.

3 Results and Discussion

Table 1 highlights our architectural enhancements.

Throughput & Reliability: Async goroutines eradicate API timeouts, enabling >5,000 concurrent mints while maintaining immediate UI responsiveness. Background polling gracefully handles network congestion via automated retries.

Privacy & Verification: The Encrypt-then-Pin pipeline resolves the transparency-privacy paradox, allowing instant on-chain SBT validation while limiting off-chain decryption strictly to authorized verifiers.

Trade-offs: Zero-friction adoption necessitates admin gas sponsorship and centralized key management. This successfully abstracts cryptography via OAuth, but shifts operational and financial burdens to the issuing enterprise.

Table 1: Traditional Web3 vs. Blockdrive System

Attribute	Traditional Web3	Blockdrive (Our)
Execution	Synchronous (RPC)	Async (Go Queue)
Throughput	~100 certs/request	>5,000 certs/batch
Data Privacy	On-chain PII risks	Encrypt-then-Pin IPFS
UX / Auth	MetaMask required	Shadow User (OAuth)
Standard	ERC-721 (Transferable)	ERC-5192 (SBT)

References

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